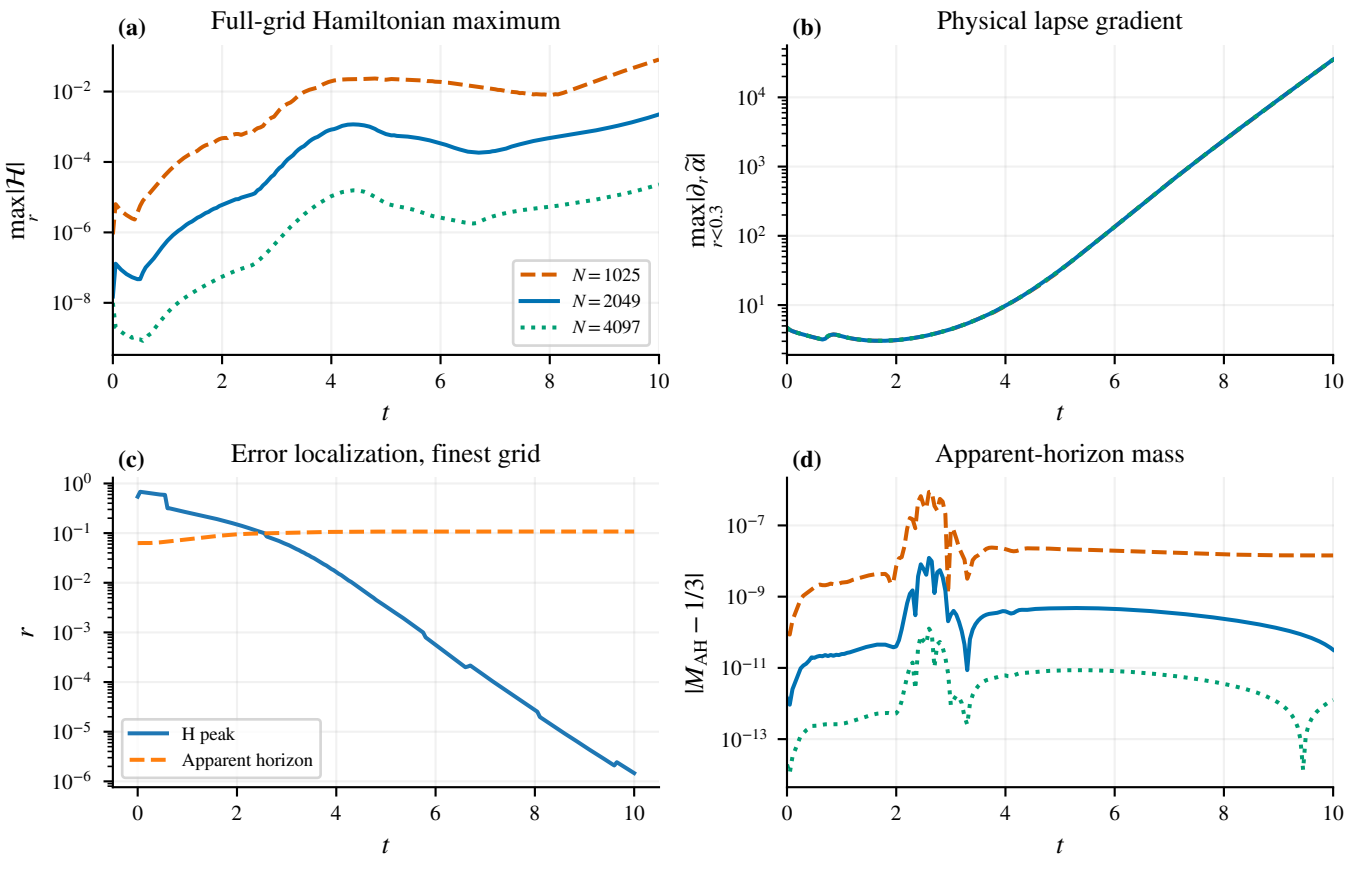
(a)
Full-grid Hamiltonian maximum
$\max_r |\mathcal{H}|$
$N = 1025$
$N = 2049$
$N = 4097$
t
(b)
Physical lapse gradient
$\max_{r<0.3} |\partial_r \tilde{\alpha}|$
t
(c)
Error localization, finest grid
r
H peak
Apparent horizon
t
(d)
Apparent-horizon mass
$|M_{\mathrm{AH}} - 1/3|$
t